1. Pancreatic lipase with Co-lipase

Pancreatic lipase can hydrolyse the ester linkages at the 1st and 3rd carbon atoms of glycerol forming 2-monoacylglycerol. The fatty acid in the second position is shifted to 1st position by an isomerase, the resulting 1 monoacylglycerol is then hydrolysed by the lipase to form glycerol and fatty acid. Co-lipase is a protein that binds to the triacylglycerol molecules at the interface, and this is essential for the action of lipase. The co-lipase is also secreted by the pancreas as zymogen, which is activated by trypsin (Figure 2.20).

Cholesterol esters are hydrolysed to free cholesterol and fatty acid by cholesterol esterase.

The phospholipid phosphatidyl choline is the second most abundant lipid in the intestinal lumen, after triglyceride. Phosphatidyl choline and cholesterol enter the gut via bile and it contributes about three fourths of the gut lipid content. Lingual and gastric lipases are ineffective in hydrolyzing phosphatidyl choline. Pancreatic phospholipase A2 (PLA2) and other lipases secreted act upon the phospholipids. The action of phospholipase A2 produces lysophospholipid and a fatty acid.

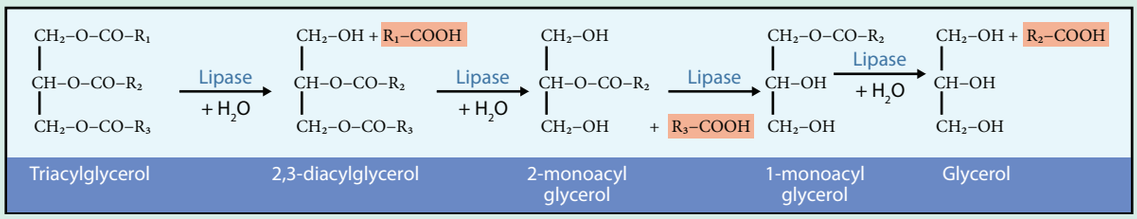


Figure 2.21 Complete hydrolysis of triglycerides

Absorption of lipids

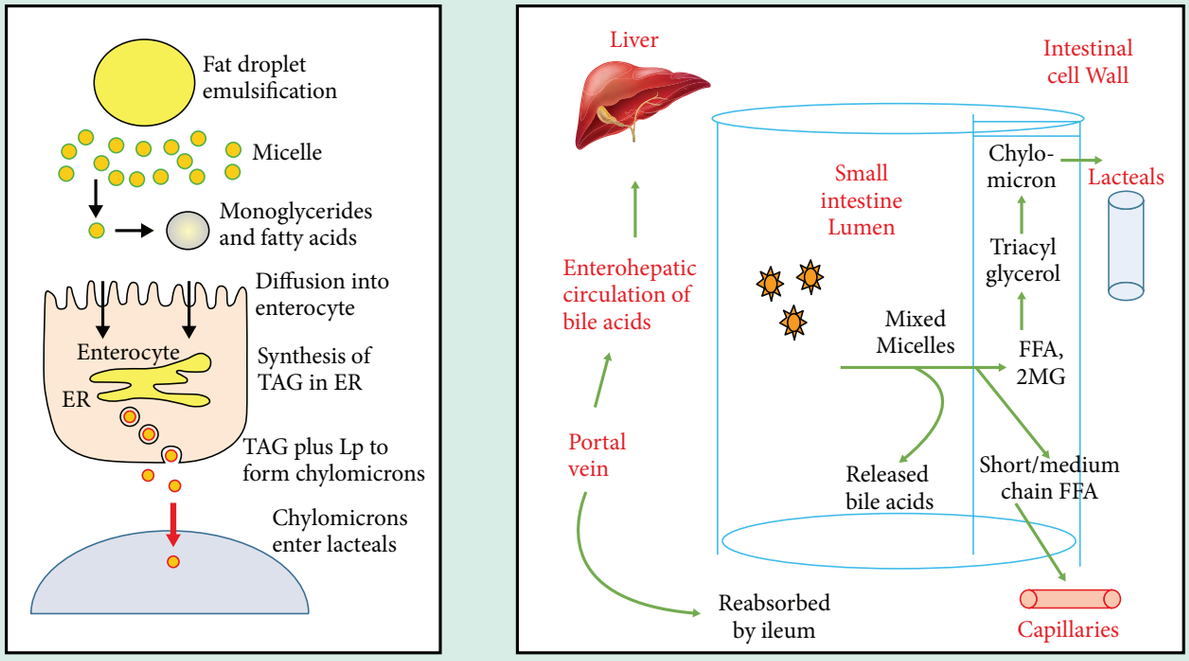


Figure 2.22 Absorption of fat as a chylomicron

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